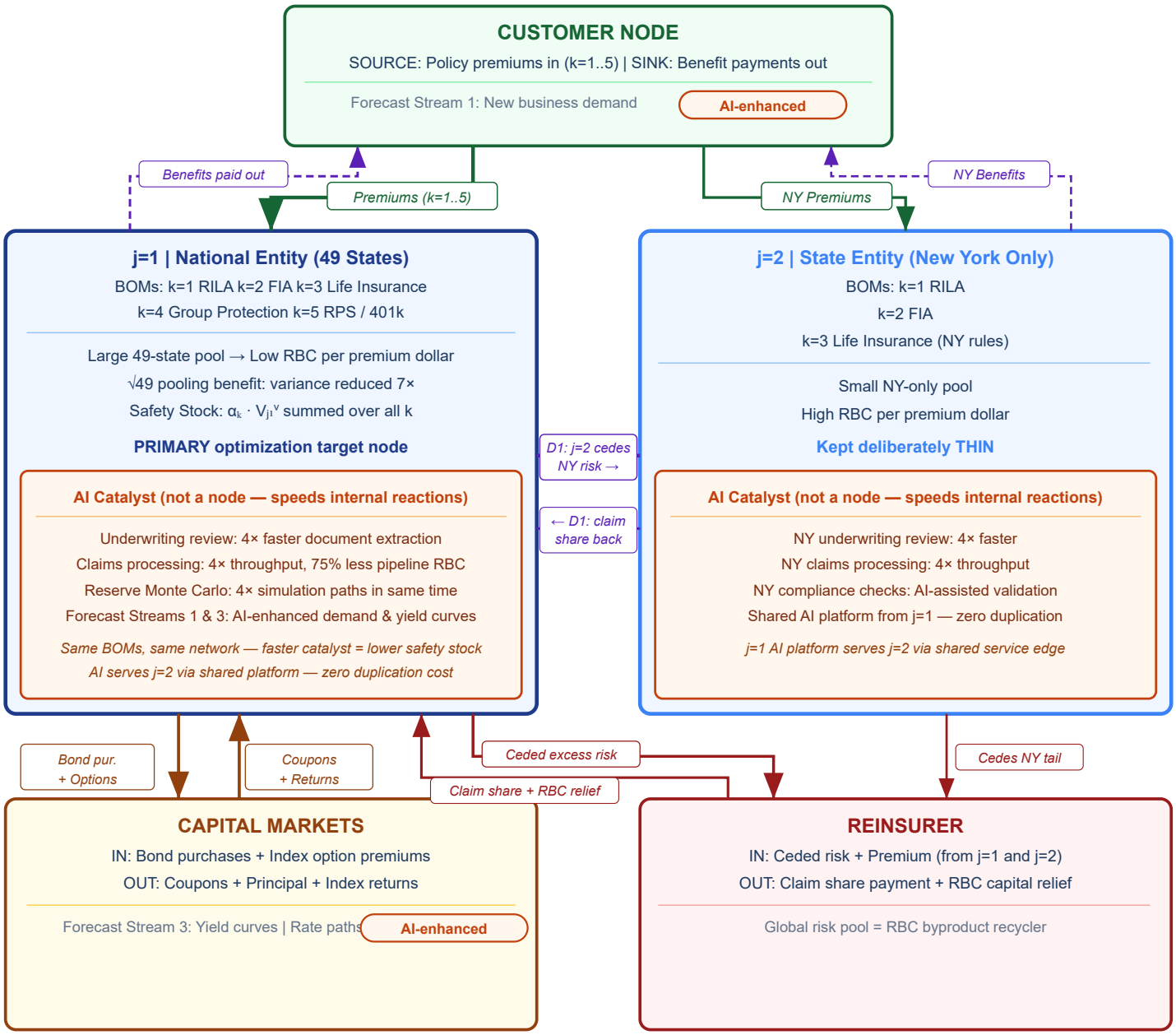
Money Flow Network Optimization

5 Nodes • 5 Product BOMs • Objective: Minimize RBC Safety Stock Across 2 Jurisdictions



NETWORK OPTIMIZATION

$$\min \sum_j \sum_k \alpha_k \cdot V_{j,k} \quad \text{s.t.} \quad \text{RBC Ratio} \geq 200\% \;|\; \text{Forecast Streams 1 \& 3} \;|\; \text{Cession limits} \;|\; \text{ERISA } (k=5 \rightarrow \text{Separate Account only})$$

Lever 1: Pool

- Route max volume through j=1
- 49-state risk diversification
- RBC per dollar → minimum
- √49 = 7× variance reduction

Lever 2: Cede

- Max D1 cession: j=2 → j=1
- Keep j=2 thin (NY law minimum)
- Cede tail risk to Reinsurer
- Reinsurer recycles RBC byproduct

Lever 3: Mix BOMs

- Shift to low-α_k products
- k=5 RPS/401k: $\alpha_k \approx 0$ (zero RBC)
- k=4 Group: α_k small
- (repriced annually, low tail)

AI Catalyst

- Claims: 4× throughput
- Underwriting: 4× faster
- Reserve Monte Carlo: 4× paths
- Pipeline RBC reduced ~75%

Key insight: Same math as consolidating 49 regional warehouses → 1 national DC to minimize total safety stock. AI is the catalyst that speeds every internal reaction — reducing lead time and pipeline RBC without changing the network structure.